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Research article

Impaired cooperation and prosocial learning during trust game in behavioral variant frontotemporal dementia

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Abstract

Background: Interpersonal trust and cooperation are important aspects of prosocial behavior. Dementia can alter social cognition, but it remains unclear how these changes alter propensity to trust and willingness to cooperate with others, and whether there are differences between types of dementia.

Objective: To determine if patients with behavioral variant frontotemporal dementia (bvFTD) differ from patients with Alzheimer's disease (AD) and cognitively normal controls in trusting behaviors and learning to cooperate.

Methods: 38 bvFTD patients, 20 AD patients and 26 cognitively normal participants invested money in two separate partners to receive a proportion of the profit in return. One partner returned a fair proportion of the investment (*cooperative* partner) while the other partner did not (*selfish* partner). We also measured perspective-taking capability for patients by administering the Interpersonal Reactivity Index to their study informants.

Results: bvFTD patients invested less money on average with the cooperative partner than the AD and control groups and did not learn to cooperate. bvFTD patients also rated the cooperative partner as less trustworthy than AD and controls. Additionally, last investment for the cooperative partner was correlated with perspective-taking skills, as was mean investment for both cooperative and selfish partners, in dementia patients.

Conclusions: Our results suggest that bvFTD patients are less receptive to reciprocity. Patients with bvFTD do not learn to cooperate with a trustworthy partner even with repeated interactions receiving positive feedback, which was related to impaired perspective-taking abilities. Our findings may shed light on disrupted reinforcement learning as a pathway to impaired prosocial behavior in bvFTD.

Keywords

Alzheimer's disease, behavior, cooperation, frontotemporal dementia, perspective-taking, prosocial, reciprocity, reinforcement learning, social, social cognition, trust

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Introduction

Learning to trust and cooperate with others is a core facet of successful social interactions and sustaining prosocial behavior in interpersonal relationships. Trust and voluntary cooperation are pivotal for successful exchanges from both an individual and organizational perspective, as in economy-based societies.[1–3] Willingness to trust and reciprocate trust are considered central components of prosociality, such that the presence of these traits may suppress selfish motives and exploitation when interacting with other people.[3,4] These trust and reciprocity behaviors have previously been linked to specific neural substrates, such as functional connectivity in the frontoparietal and cingulo-opercular networks[4] and in the functional integration of the caudate, amygdala, lateral prefrontal cortex, temporo-parietal junction, and temporal poles.[5]

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Neurodegenerative diseases that involve these regions may thus impact the patients' ability or willingness to trust and voluntarily cooperate with other people, as their propensity for trust and social reward processing and decision-making processes may be disturbed through the progression of their disease. Behavioral variant frontotemporal dementia (bvFTD) is known for marked changes in personality and behavior, including loss in empathy, emotion processing, and perspective-taking capabilities.[6–8] Socio-cognitive abilities like perspective-taking, which are disrupted in bvFTD, have been related to trust and reciprocity in healthy adults and adolescents,[9,10] but it remains unknown whether impaired perspective-taking is also associated with impaired trust and reciprocity in patients with dementia. Prior studies have also linked decreased trust propensity with cognitive impairment,[11,12] hence motivating the present study to investigate trusting behaviors in bvFTD as well as Alzheimer's disease (AD) patients.

In neuroeconomics, the Trust Game[13] is a frequently administered experimental paradigm for measuring behavioral trust and trustworthiness. Participants playing the classic Trust Game are designated the role as an investor who is given a sum of money and decides how much to give to the other player after being told that the investment will be multiplied by some factor (e.g., by a factor of 4) to arrive at a total profit. The recipient then returns some proportion of the profit back to the investor, which can be either fair (i.e., a larger amount than the initial investment) or unfair (i.e., less than the initial investment). Trust is measured as the willingness of the investor to give money to the other player and is reinforced by a pattern of fair return of the profits. When played over multiple rounds, the task also provides information about reinforcement learning as participants progressively acquire information on whether their choice to invest with a given player is rewarded with fair offers or punished with unfair offers. Versions of the Trust Game have been implemented to study differences in trusting behavior in younger versus older adults[14] as well as clinical populations, including bvFTD and AD,[15] Parkinson's disease[16] and major depressive disorder.[17] However, prior trust studies in dementia populations have not yet shown group differences in investment behaviors over the course of the game or how the behaviors may relate to impaired perspective-taking.

The present study compares trust and investment behavior in bvFTD patients, AD patients, and cognitively normal controls during a repeated Trust Game and further examines whether these behaviors differ based on opponent type (cooperative versus selfish). We then test the relationship between the Trust Game measures and perspective-taking ability, as indexed by the Interpersonal Reactivity Index.[18]

Methods

Participants

Patients were recruited from the Behavioral and Cognitive Neurology clinic in the Vanderbilt University Medical Center between July 2018 and November 2023. For all study participants, written informed consent was obtained from each participant as well as their study informant. Study protocol was approved by the Vanderbilt Institutional Review Board. Record review by a behavioral neurologist (RRD) confirmed the diagnosis for each patient included in the study in accordance with international research criteria.[19–21] Inclusion criteria for patients required a diagnosis of bvFTD or AD. Caregivers and study informants of patients were recruited as healthy controls to participate in this study. Dementia severity was rated by the study neurologist as mild cognitive impairment (MCI), mild dementia, or moderate dementia. Patients at the MCI stage were permitted if they had clinical and biomarker evidence consistent with a specific underlying neurodegenerative disease (e.g., abnormal PET scan).

Demographic information was collected for all patients, including age, sex, race/ethnicity, education background (asked highest degree or level of school completed – see each stratification in Table 1), and informant relationship. Study inclusion required that patients have a reliable study informant present for the study visit, defined as any person who spends a significant period of time with the patient, i.e., in-person interactions greater than or equal to two times per week. Applying these criteria resulted in 38 bvFTD and 20 AD patients along with 26 cognitively normal controls being included in the current analysis. The healthy control participants in this study were primarily caregivers of the enrolled patients: while they completed the Trust Game task, additional assessments were limited. The Montreal Cognitive Assessment (MoCA) was administered to all patients (but not to healthy controls) as a generalized measure of cognitive function.[22] Additionally, because study informants were not available for the control participants, they did not complete the IRI.

Experimental procedure

All participants completed a modified version of the Trust Game.[13] Instructions were presented on a laptop screen, stating that each participant will play as an investor with two different persons for 10 consecutive rounds with each partner (Figure 1). They were told that the objective of the game is to maximize their own winnings while playing with each partner. Participants received $10 on each round (visualized on screen) and told they could transfer any amount from $0 to $10 to the other player. Then, the

Table 1. Demographic characteristics of sample.

	AD	bvFTD	HC	
	(N = 20)	(N = 38)	(N = 26)[a]	Statistical Analysis
Age				
Mean (SD)	69.6 (8.79)	62.8 (9.34)	63.2 (8.72)	AD > bvFTD ($p = 0.02$)*[b]
Sex				
F (%)	8 (40.0%)	6 (15.8%)	17 (65.4%)	bvFTD > HC ($p < 0.001$)*[c]
M (%)	12 (60.0%)	32 (84.2%)	9 (34.6%)	
MoCA				
Mean (SD)	18.6 (5.33)	20.3 (5.40)	–	$p = 0.26$ [b]
Disease severity				$p = 0.059$ [d]
Mean (SD)	0.950 (0.426)	1.20 (0.514)	–	
Education				
High school degree (%)	4 (20%)	7 (18.4%)	2 (13.3%)	$p = 0.807$ [e]
Some college (%)	5 (25%)	4 (10.5%)	1 (6.7%)	
Associate degree (%)	1 (5%)	6 (15.8%)	4 (26.7%)	
Bachelor's degree (%)	6 (30%)	11 (28.9%)	4 (26.7%)	
Master's degree (%)	2 (10%)	7 (18.4%)	3 (20%)	
Professional degree (%)	2 (10%)	2 (5.3%)	1 (6.7%)	
Doctorate (%)	0 (0%)	1 (2.6%)	0 (0%)	
Informant relationship				
Child (%)	2 (10.0%)	4 (10.5%)	–	$p = 0.203$ [e]
Friend (%)	1 (5.0%)	0 (0%)	–	
Other (%)	1 (5.0%)	1 (2.6%)	1 (12.5%)	
Spouse (%)	16 (80.0%)	33 (86.8%)	7 (87.5%)	
Race				
White (%)	20 (100%)	36 (94.7%)	14 (93.3%)	$p = 0.539$ [e]
African-American (%)	0 (0%)	2 (5.3%)	1 (6.7%)	

AD: Alzheimer's disease; bvFTD: behavioral variant frontotemporal dementia; HC: healthy controls; SD: standard deviation; MoCA: Montreal Cognitive Assessment.
*$p < 0.05$.
[a]Demographic information and behavioral scores were limited for healthy control subjects:
MoCA scores were only available for 5 controls.
Education and race measures were only available for 15 controls.
Informant relationship was only available for 8 controls.
[b]One-way ANCOVA test.
[c]Fisher's exact test.
[d]Kruskal-Wallis test.
[e]Chi-square test.

experimenter would quadruple (x4) this transferred amount, so that the other player receives four times the number of dollars the participant transferred. After each investment decision, the other player could transfer back to the participant any amount of the quadrupled number of dollar bills that they received.

In this Trust Game, participants played against two different partners in a fixed order. At the beginning of each partner block, participants were shown a photograph of the partner's face on screen and asked to rate how trustworthy they believed the person to be using a scale from 0 to 10, where 0 = "Not at all trustworthy" and 10 = "Extremely trustworthy" (*pre-rating*). Then, they played 10 rounds against that partner. In each round, participants were given $10 and asked how much in dollars they would like to invest in the partner. After choosing an amount, participants were shown how much the partner received (4× the investment amount) and how much the partner returned to them, displaying both players' earnings. After completing 10 trials with the first partner, participants were reminded of their total earnings with that partner and were again asked to rate the partner's trustworthiness using the same 0–10 scale (*post-rating*). This full procedure was then repeated with the second partner.

The experimental design for this modified version of the Trust Game involved all participants playing first with a *cooperative* (fair) partner, followed by a *selfish* (unfair) partner. The order of partners was fixed across participants to reduce task complexity and cognitive load, eliminating the potential need for reversal learning. By keeping partner order consistent and using fixed patterns of behavior (i.e., each partner always returned either a fair or unfair proportion of the investment), the task isolated trust-based learning rather than testing participants' ability to detect

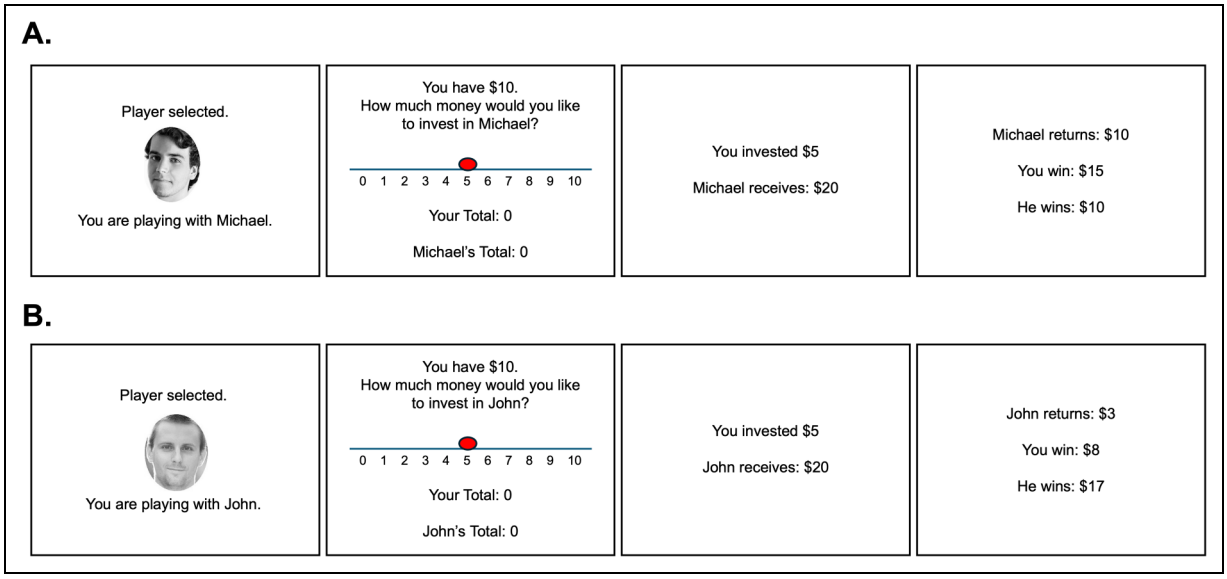

Figure 1. Example trials for the (A) cooperative partner and the (B) selfish partner for modified trust game task design. Participant selects amount of money out of $10 to invest in partner, amount is multiplied by 4, and portion of the 4× investment is returned.

contingency switches. The return on investment from each partner was randomized within a defined range: the *cooperative* partner returned between 37.5 and 50% of the participant's investment amount (a net benefit of 1.5–2×), while the *selfish* partner returned between 12.5 and 25% of the investment (a net loss of 0–0.5×). Returns were randomized within those ranges to maintain ecological validity while ensuring consistency in reinforcement type.

For this study, mean investment (average investment amount over 10 trials) and last investment (dollar amount invested on the last trial) for the cooperative and the selfish partner were operationalized as *trusting behavior*. Change in investment (last minus first investment) was operationalized as *learning* and trustworthy ratings (pre-rating, post-rating) were operationalized as *trust perception*.

Behavioral measures

The *Interpersonal Reactivity Index* (IRI) is a commonly used self-report questionnaire that measures empathy across four subscales: Perspective-Taking, Fantasy, Empathic Concern, and Personal Distress.[18] Each of the four subscales is comprised of seven items answered using a 5-point Likert scale (0 = "Does not describe me well" to 4 = "Describes me very well"), resulting in a score range of 0 to 28 per subscale. For this study, we used the Perspective-Taking subscale (IRI-PT) to gauge one's ability to adopt the viewpoint of others. Higher scores on the IRI-PT indicate greater perspective-taking ability. An example item is "I try to look at everybody's side of a disagreement before I make a decision." As has been done in prior studies of patients with neurodegenerative disorders, we administered the IRI questionnaire to the study informants of the patients due to concerns of patients being unaware of their socioemotional deficits.[23–25] The IRI-PT subscale has been used in prior studies linking perspective-taking with trust and prosocial behavior[26–29] and has been applied to show cognitive empathy is significantly impaired in patients with bvFTD compared to both individuals with AD and healthy controls.[8,23]

Statistical analysis

Demographics and participant characteristics were summarized across the three groups using means with standard deviation or sample proportions, where appropriate. ANOVA was used to compare continuous variables, Fisher's exact test was used for categorical variables, and Kruskal-Wallis test was used for ordinal variables. We used linear regression to test group differences in mean, last, first, and change in investment across the two opponent types by modeling the main effects and interactions of diagnostic group and opponent type controlling for age and sex. Within these models, we tested hypothesized differences between investment amounts between groups for each opponent type, using t-tests adjusting for six separate tests (three group comparisons for each opponent type) with Tukey's adjustment.[30] We used robust standard errors so that the comparisons were not sensitive to unknown heteroskedasticity.[31] The same models were used to compare opponent pre- and post-rating between groups. We visualized the results using estimated marginal means. Finally, to study the association between investment measures and opponent trustworthiness, we fit linear regression models with investment measures as the outcome and opponent trustworthiness, age, and sex as independent variables.

The same model was also fit with perspective-taking ability instead of opponent trustworthiness as the independent variable for only the patient groups, since the IRI measure was not available for controls.

Results

Participant characteristics

38 bvFTD and 20 AD patients as well as 26 cognitively unimpaired controls were included in this study (Table 1). The bvFTD group was younger than the AD group (62.8 ± 9.3 versus 69.6 ± 8.8; $F[2,81] = 4.07$, $p = 0.02$) but not younger than the controls group (63.2 ± 8.7) in a one-way ANOVA test. bvFTD patients had a higher proportion of males than controls (84% versus 35%, Fisher's exact test: $p < 0.001$) but not the AD group ($p = 0.17$, 60% male). Mean MoCA scores for the bvFTD and AD patients did not significantly differ (20.3 ± 5.4 versus 19 ± 5.2, $F[1,56] = 1.29$, $p = 0.26$) in a one-way ANCOVA test between the two patient groups. Dementia severity ratings were not significantly different between bvFTD and AD patients (Kruskal-Wallis test: $\chi^2 = 3.57$, $p = 0.059$). Some demographic variables, such as race, education, and informant relationship, have missing data for a subset of healthy control participants due to limited data collection procedures early in the study. As a result, the total number of observations may be less than 26 for certain variables (see Table 1 for exact subset numbers). Fisher's exact test was performed for race, education and informant relationship proportions between the bvFTD, AD and control subset groups and no significant group differences were found (Table 1).

Trust game investment behavior group differences

Group differences in Trust Game investment behaviors were determined using linear regression models. Overall, bvFTD patients invested less money in their partners than AD patients and healthy controls (see Table 2 for all following test statistics and p-values). For the cooperative partner, mean investment was significantly lower in the bvFTD group (mean 4.78 ± 2.68) compared to AD (6.12 ± 1.98) and controls (6.5 ± 2.04). For the selfish partner, mean investment did not significantly differ between the bvFTD (3.73 ± 2.4) and AD (4.78 ± 1.83) groups or between bvFTD and controls (4.18 ± 2.03). Mean investment did not significantly differ between the AD and controls groups for either the cooperative partner or the selfish partner. Last investment (the investment amount on the last trial of each round) was significantly lower in the bvFTD (4.71 ± 3.14) group compared to AD (6.95 ± 2.56) and controls (6.54 ± 2.49) for the cooperative partner, but not for the selfish partner for bvFTD (3.05 ± 2.78) and either AD (4.1 ± 2.36) or controls (3.08 ± 2.76). Last investment did not significantly differ between AD and controls for either the cooperative or selfish partner. First investment (the investment amount on the first trial of each round) did not significantly differ for the cooperative partner between bvFTD (5.16 ± 2.97) and AD (5.05 ± 2.21) or bvFTD and controls (5.31 ± 2.22) or for the selfish partner between bvFTD (5.13 ± 3.12) and AD (5.25 ± 2.15) or bvFTD and controls (6.0 ± 2.45). First investment did not significantly differ between AD and controls for either the cooperative or selfish partner.

Cooperative learning behavior group differences

Change in investment (calculated as the last investment minus the first investment) was operationalized to measure a participant's ability to learn whether to trust either partner based on feedback from prior trials. All three groups learned to not trust the selfish partner – defined as a significant reduction in the investment amount between the first and last trials (Figure 2). There were no significant differences in change in investment with the selfish partner between patients with bvFTD (−2.08 ± 3.25) and AD (−1.15 ± 2.28), between bvFTD and controls (−2.92 ± 3.4), or between AD and controls (Table 2). However, both the AD (1.9 ± 2.49) and controls (1.23 ± 2.01) groups did change their investment behavior while playing with the cooperative partner by increasing their investment amounts over time (Figure 2). In contrast, the bvFTD group (−0.45 ± 3.13) differed from the other two groups by investing less money over time with the cooperative partner, compared to AD and controls (Table 2).

Trustworthiness rating group differences

Post-ratings of trust for the cooperative partner were significantly lower in the bvFTD group than the AD ($t = 2.69$, $p = 0.02$) and control groups ($t = 2.64$, $p = 0.02$), while post-ratings did not differ for the selfish partner between the three groups (Table 3). Pre-ratings of trust did not differ for either the cooperative or selfish partner between the three groups (Table 3). Change in rating (calculated as post-rating minus the pre-rating for each opponent) was significantly different between the bvFTD and AD groups for the cooperative partner only ($t = 3.19$, $p = 0.005$) but not for the selfish partner ($t = 0.40$, $p = 0.92$; Figure 3). Change in rating did not significantly differ between the bvFTD and control groups for either the cooperative partner ($t = 2.09$, $p = 0.10$) or the selfish partner ($t = 1.85$, $p = 0.16$; Table 3).

Association between investment behavior and ratings of trustworthiness

Change in investment (last minus first investment) was significantly related to change in trust rating (post-rating minus pre-rating) for the cooperative partner for the combined

Table 2. Pairwise tests for group differences in investment. p-values are adjusted using Tukey's method within each investment type category.

Investment Type	Opponent	Contrast	Estimate	SE	T-statistic	*p*
First	Cooperative	AD - bvFTD	0.155835	0.71819	0.216983	0.974383
		AD - HC	−0.14656	0.659939	−0.22208	0.973182
		bvFTD - HC	−0.30239	0.695077	−0.43505	0.901009
	Selfish	AD - bvFTD	0.382151	0.741543	0.515345	0.86398
		AD - HC	−0.63887	0.671033	−0.95207	0.608207
		bvFTD - HC	−1.02102	0.716186	−1.42563	0.330171
Mean	Cooperative	**AD - bvFTD**	**1.627727**	**0.60444**	**2.692951**	**0.021273**
		AD - HC	−0.33203	0.584208	−0.56835	0.837135
		bvFTD - HC	**−1.95976**	**0.611048**	**−3.20721**	**0.00459**
	Selfish	AD - bvFTD	1.332464	0.57813	2.304782	0.057994
		AD - HC	0.639505	0.562239	1.137425	0.492528
		bvFTD - HC	−0.69296	0.569287	−1.21724	0.444743
Last	Cooperative	**AD - bvFTD**	**2.356789**	**0.76027**	**3.099935**	**0.006439**
		AD - HC	0.429822	0.784651	0.547788	0.847757
		bvFTD - HC	**−1.92697**	**0.745476**	**−2.58488**	**0.028514**
	Selfish	AD - bvFTD	1.164684	0.697339	1.670183	0.219848
		AD - HC	1.04136	0.786145	1.324643	0.383608
		bvFTD - HC	−0.12332	0.747235	−0.16504	0.985097
Change	Cooperative	**AD - bvFTD**	**2.200954**	**0.827737**	**2.659003**	**0.02335**
		AD - HC	0.576382	0.677045	0.851321	0.671681
		bvFTD - HC	−1.62457	0.693937	−2.34109	0.053111
	Selfish	AD - bvFTD	0.782533	0.761659	1.027406	0.560695
		AD - HC	1.680228	0.807504	2.080767	0.097063
		bvFTD - HC	0.897695	0.886948	1.012117	0.570307

Significant comparisons ($p < 0.05$) marked in bold. AD: Alzheimer's disease; bvFTD: behavioral variant frontotemporal dementia; HC: healthy controls; SE: standard error.

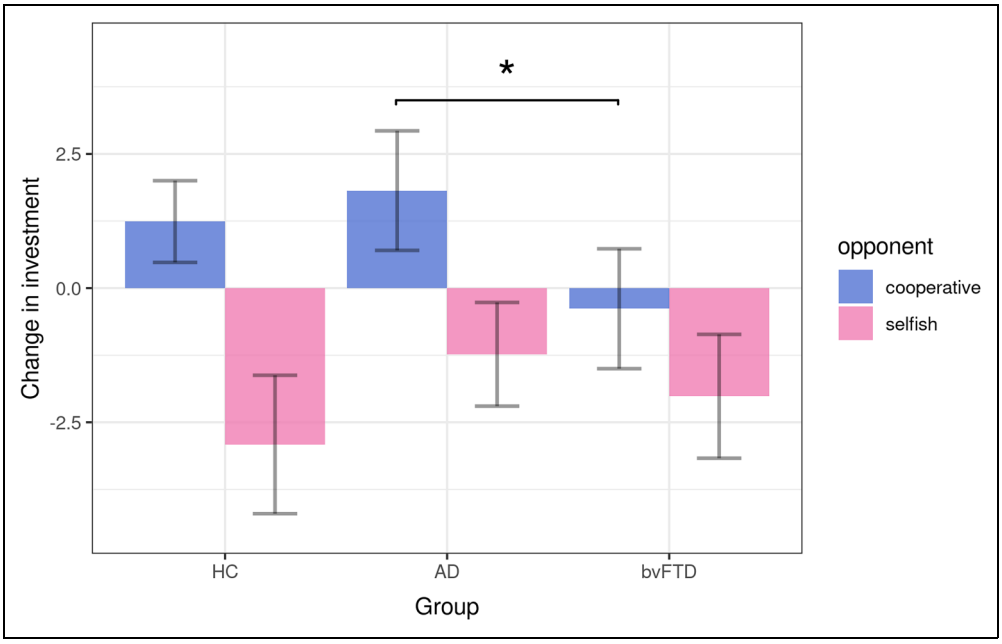

Figure 2. Change in investment (last investment minus first investment) was significantly lower in patients with bvFTD compared to patients with AD for the cooperative partner, but not for the selfish partner. *p < 0.05.

group of bvFTD and AD patients ($\beta = 0.49$, $t = 3.51$, $df = 54$, $p < 0.001$) after controlling for age and sex. Change in investment was also significantly related to change in trust rating for the selfish partner, for the combined group ($\beta = 0.68$, $t = 4.90$, $df = 54$, $p < 0.001$). Change in investment was significantly related to change in trust rating for

Table 3. Pairwise tests for group differences in rating. p-values are adjusted using Tukey's method within each rating type category.

Rating Type	Opponent	Contrast	Estimate	SE	T-statistic	*p*
Prerating	Cooperative	AD - bvFTD	−0.5362	0.647486	−0.82812	0.68615
		AD - HC	−1.01454	0.487629	−2.08056	0.097107
		bvFTD - HC	−0.47834	0.620253	−0.77121	0.721208
	Selfish	AD - bvFTD	0.695381	0.657909	1.056956	0.542194
		AD - HC	−0.10685	0.53662	−0.19912	0.978383
		bvFTD - HC	−0.80223	0.595937	−1.34617	0.371867
Postrating	Cooperative	**AD - bvFTD**	**1.989459**	**0.738754**	**2.692994**	**0.02127**
		AD - HC	0.255967	0.590998	0.43311	0.901845
		bvFTD - HC	**−1.73349**	**0.655844**	**−2.64315**	**0.02438**
	Selfish	AD - bvFTD	1.04209	0.967527	1.077066	0.529672
		AD - HC	1.679044	0.822212	2.042106	0.105572
		bvFTD - HC	0.636953	0.699753	0.910254	0.634626
Change	Cooperative	**AD - bvFTD**	**2.525657**	**0.791644**	**3.190396**	**0.004843**
		AD - HC	1.270508	0.643379	1.974742	0.121796
		bvFTD - HC	−1.25515	0.601861	−2.08544	0.096072
	Selfish	AD - bvFTD	0.34671	0.867325	0.399746	0.915741
		AD - HC	**1.785893**	**0.714624**	**2.499065**	**0.035715**
		bvFTD - HC	1.439183	0.780087	1.8449	0.15844

Significant comparisons ($p < 0.05$) marked in bold. AD: Alzheimer's disease; bvFTD: behavioral variant frontotemporal dementia; HC: healthy controls; SE: standard error.

the selfish partner for the control group ($\beta = 0.37$, t = 2.64, df = 22, p = 0.015) but did not reach significance for the cooperative partner ($\beta = 0.24$, t = 1.78, df = 22, p = 0.089). Change in investment was significantly related to change in trust rating for the selfish partner for both bvFTD and AD patients (bvFTD: $\beta = 0.69$, t = 3.96, df = 34, p < 0.001; AD: $\beta = 0.76$, t = 3.76, df = 16, p = 0.002) after controlling for age and sex. There was a trend towards change in investment being positively associated with change in trust rating for the cooperative partner for bvFTD and AD (bvFTD: $\beta = 0.32$, t = 2.01, df = 34, p = 0.051; AD: $\beta = 0.60$, t = 2.00, df = 16, p = 0.064).

Association between investment behavior and impaired perspective-taking abilities

IRI scores were only available for the bvFTD and AD patient groups and not for the healthy controls, as the IRI questionnaire was only administered to study informants of the patients. For the combined patient groups, IRI perspective-taking scores were significantly related to mean investment ($\beta = 0.17$, t = 3.03, df = 44, p = 0.004) and last investment ($\beta = 0.18$, t = 2.64, df = 44, p = 0.011) for the cooperative partner after controlling for age and sex and was significantly related to mean investment for the selfish partner ($\beta = 0.15$, t = 2.56, df = 44, p = 0.013). For the bvFTD patient group, IRI perspective-taking scores were significantly related to mean and last investment (mean: $\beta = 0.23$, t = 2.20, df = 27, p = 0.04; last: $\beta = 0.27$, t = 2.37, df = 27, p = 0.025) for the cooperative partner after controlling for age and sex.

Discussion

Our results showed that patients with bvFTD cooperated less during the Trust Game compared to patients with AD and cognitively unimpaired healthy controls. While playing with the selfish partner, the participant groups did not differ in their investment behavior over repeated interactions, as all groups learned to invest less in the selfish player over time. In contrast, while playing with the cooperative partner, only the AD and control groups learned to trust by investing more money over repeated trials, whereas the bvFTD group did not learn to trust the cooperative partner and invested less money by the end of the game. Our findings further showed that reduced investment with the cooperative partner in bvFTD was associated with reduced trustworthiness ratings for that partner. Additionally, investment with the cooperative partner was significantly related to perspective-taking abilities in bvFTD. Overall, our results indicate that bvFTD patients are less trusting and less receptive to positive reinforcement while learning to trust than patients with AD and healthy controls.

Trust and investment behaviors in bvFTD and AD

The Trust Game has been utilized over the past few decades as a laboratory measure of willingness to trust and reciprocate trust in various participant cohorts, including adolescents, aging adults, and adults with psychiatric illness.[2,13] Economic games like the Trust Game simulate real-world social situations ingrained with tensions between self-interest and social reciprocation. Alterations

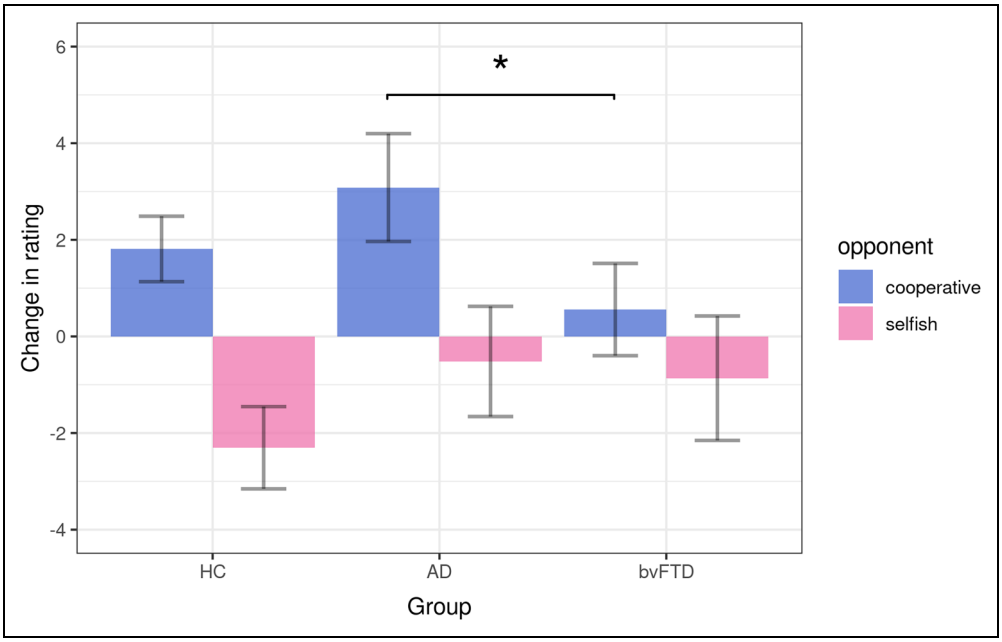

Figure 3. Change in rating (post-rating minus pre-rating) was significantly lower in patients with bvFTD compared to patients with AD for the cooperative partner, but not for the selfish partner or between the bvFTD and control groups. * $p < 0.05$.

in learning and decision-making related to trust may produce dire functional consequences as trust and trustworthiness are pivotal to successful social relationships and cooperation among individuals. However, the Trust Game has rarely been administered to assess trust and social reciprocity in bvFTD or AD patients. One prior study applied a modified Trust Game to assess learning and memory of trust behavior across AD and bvFTD patients and healthy controls.[15] Wong and colleagues administered a multi-round Trust Game comprising trustworthy and untrustworthy conditions and compared binary choices between investing or keeping the money. They reported a significant group difference in overall investment behaviors between the patients and controls but did not find a difference between bvFTD and AD patients themselves. Further, in a post-hoc analysis, Wong and colleagues found that among bvFTD patients, there was an asymmetry in learning, with greater accuracy in choices in the untrustworthy condition than the trustworthy condition, in contrast with only a trend in this direction in AD patients and no evidence of asymmetry in controls. This aligns with the results of our study in which AD and control groups learned to trust the cooperative partner while bvFTD patients did not, indicating that this lack of cooperation may differentially emerge in bvFTD. Given that learning to cooperate with unfamiliar people is an important aspect of prosocial behavior,[32] the inability of individuals with bvFTD to do so may contribute to their problematic changes in social behavior. The relative lack of significant differences in trusting behaviors between AD patients and healthy controls further reinforce this hypothesis, as individuals with early- or mild-stage AD often retain basic social decision-making abilities, particularly in structured paradigms that provide consistent feedback and limited cognitive demand like the TG.[15,53]

Impaired prosocial learning in bvFTD

The selective divergence in investment behavior in the trustworthy condition, while showing normal investing in the untrustworthy condition, suggests that bvFTD patients were more influenced by negative reinforcement than positive reinforcement during the game. Thus, one potential interpretation for our findings is that bvFTD patients have selectively impaired positive reinforcement learning, leading to reduced cooperative behavior.

Much of the prior literature on altered reward and reinforcement processing in bvFTD has investigated the underlying changes in bvFTD patients' pursuit or valuation of rewards and provided evidence of reduced reward responsiveness/sensitivity to monetary reward and loss as well as aberrant hedonic valuation and reward prediction.[33–37] Prior studies have observed that bvFTD symptoms of overeating, craving of sweet foods, hypersexuality, hyposexuality, and risky decision-making may be tied to altered sensory processing or shifts in the perception of specific types of reward.[36–40] Evidence for reduced sensitivity to aversive stimuli, such as unpleasant odors, in bvFTD has also been reported, suggesting that insensitivity to negatively valenced information might contribute to behavioral choices in bvFTD.[39,41] This is in contrast to the results of the current study, where

bvFTD patients learned to avoid investing with the selfish partner via negative reinforcement, but did not learn to invest in the cooperative partner through positive reinforcement. It is worth noting that although bvFTD patients have impaired behavioral flexibility and reversal learning,[42–47] those deficits cannot explain our results since in our Trust Game paradigm, participants always played with the cooperative partner first and both the cooperative and selfish partners always provided consistent returns on investments. Importantly, while both bvFTD and AD are associated with progressive cognitive decline, the group differences we observed in trust behavior with the cooperative partner are unlikely to be solely attributable to general cognitive impairment, as in our patient sample of mild-to-moderate dementia, we did not find significant differences in global cognitive function (MoCA) or dementia severity ratings. As such, bvFTD patients' observed impairments in learning to trust the cooperative partner, compared to AD and healthy controls, are more likely to reflect selective disruption of socioemotional or reward-based processing mechanisms rather than differences in general cognitive capacity.

It is also possible that the lack of learning to trust in bvFTD was influenced by the specific reward structure in the task. While the cooperative partner's returns (37.5–50% of the investment) were intended to represent a moderately generous and socially plausible partner, this return range may not have been sufficiently salient to drive prosocial learning in bvFTD. Prior research suggests that individuals with bvFTD exhibit blunted sensitivity to reward and may require higher reward magnitude to appropriately update expected value or adjust behavior.[48] In contrast, the same feedback paradigm appeared sufficient to elicit adaptive investment behavior in both the AD and control groups. These findings suggest that bvFTD patients may require stronger or more explicit reinforcement signals to guide prosocial learning, and future studies should explore whether increasing the magnitude or salience of positive social feedback could improve trust-related behavior in this population.

Decreased trust and reinforcement learning underlies investment behavior in bvFTD

A comprehensive neuropsychological model of trust remains opaque, as multiple cognitive and social processes may underlie trust propensity and perception of trustworthiness, including understanding of reciprocity, or the social principle of give and take. In this context, the learning of reciprocity is inherently tied to trust and trusting behaviors. Here, we found parallel alterations in positive investment learning and subjective ratings of trustworthiness among bvFTD patients based on their partners' reciprocity. A few prior studies have investigated the possibility that socially reinforced learning, or learning to cooperate with others following social feedback, is disrupted in bvFTD and have found that impaired social cognitive abilities in bvFTD patients may prevent the incorporation of social information while making decisions and disturb their capability for reinforcement learning.[49,50] Our findings expand upon these prior reinforcement learning findings by showing that specifically *positive* social reinforcement learning appears disrupted in bvFTD while negative reinforcement learning may remain intact. This is interesting considering that much of the literature on emotional reactivity in bvFTD has found that patients with bvFTD showed impaired recognition of negative but not positive facial emotion expressions and reactivity.[41,51,52] Further research testing associations between emotion recognition and trust behavior will be useful in resolving these different patterns of results.

Impaired perspective-taking underlying investment behavior in bvFTD

In our study, lower mean and last investment amounts were associated with impaired perspective-taking in the bvFTD group (as rated by their study informants). The ability to adopt another person's perspective and motivations, i.e., theory of mind, may be critical to socially reinforced learning, as reciprocal behavior requires understanding and reasoning about others' intentions and beliefs,[10] and both of these processes are disturbed in bvFTD patients.[50] Deciding to trust may be a learned prosocial behavior requiring the ability to infer a stranger's positive intentions based on their prior investment behavior. Our results are in line with prior studies that have linked higher IRI-PT scores with greater interpersonal trust, higher investment in cooperative or trustworthy partners, and greater responsiveness to social feedback in both adolescents and adults.[26–29] For example, adolescents with higher self-reported PT were more likely to invest in a trustworthy partner over repeated rounds of a Trust Game, and perspective-taking moderated the development of reciprocity and cooperations.[9] These findings support utilizing the IRI-PT as meaningful measure of cognitive empathy/theory of mind and suggest that specific impairments in perspective-taking in bvFTD contribute to reduced adaptive investment and learning to trust. bvFTD patients may lack the ability to make these inferences regarding positive intentions of cooperative partners, impeding their ability to learn to invest more in trustworthy partners over time.

Limitations

Our study has several important limitations. First, our sample size was relatively limited, and was restricted to a single center with a homogeneous racial and ethnic

patient population. Multi-center replication consisting of a larger and more diverse patient sample is needed. One important limitation of our patient population is that our sample of AD patients is restricted to mostly mild dementia patients and does not represent the full breadth of AD severity. Future studies would benefit from inclusion of more moderately severe AD patients in order to fully characterize the trusting differences between AD and bvFTD patients as AD cases become more severe. We did not collect a specific measure of disease duration and could therefore not control for disease duration as a covariate in our analyses. We also did not collect comprehensive measures for all the participants in our study, such as IRI scores for the cognitively unimpaired healthy controls. Another limitation of our current study is that disease-related symptoms like apathy could potentially influence our results, and we did not have a direct measure of apathy available for these analyses. We plan to address this in future studies as we have begun collecting the Neuropsychiatric Inventory, which includes a subscale for apathy, for all participants. Future studies could benefit from matching healthy controls with patient populations in demographic characteristics such as age, years of education, or estimated premorbid cognitive functioning of patients. However, while better matching is important in considering findings related to controls, it is notable that the comparisons to AD patients suggests a level of specificity that is not simply due to a process of generically impaired memory or ability to attend to the task. Finally, our version of the economic Trust Game was limited in number of trials, number and diversity of opponents, and the order of partner presentation during the game was not randomized or counter balanced. Future studies incorporating a larger number of trials would also allow for the testing of specific computational models of reinforcement learning parameters.

Conclusions

The present study examined trust and reciprocity in bvFTD and AD patients using the Trust Game, a common quantitative behavioral economics measure. Our results indicate that bvFTD patients were less trusting with their investments with the cooperative partner compared to the AD and control groups and this reduced investment was also associated with reduced trust ratings in the cooperative partner. Lower mean and last investment in the cooperative partner were also associated with impaired perspective-taking skills in bvFTD patients. bvFTD patients appeared selectively impaired in their ability to learn to cooperate with trustworthy partners, even with repeated interactions receiving positive feedback. This deficit in social reciprocal learning may in turn lead to reduced prosocial behavior in bvFTD.

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Ethical considerations

The study was approved by the Vanderbilt University Institutional Review Board.

Consent to participate

All participants provided written informed consent.

Consent for publication

Not applicable.

Author contribution(s)

Jayden J Lee: Conceptualization; Methodology; Visualization; Writing – original draft; Writing – review & editing.
Jiangmei Xiong: Formal analysis; Visualization.
Tony X Phan: Data curation; Software.
Jerica E Reeder: Data curation.
Lindsey C Keener: Data curation.
Matthias G Tholen: Methodology; Software.
Simon Vandekar: Formal analysis; Supervision.
David H Zald: Writing – review & editing.
R Ryan Darby: Conceptualization; Methodology; Supervision; Writing – review & editing.

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability statement

The data supporting the findings of this study are available on request from the corresponding author.

Supplemental material

Supplemental material for this article is available online.

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